

SNEHAL NAIR

Head of Machine Learning | Applied AI Leadership | Remote UK/Europe

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PROFILE

ML leader with 13+ years building and scaling high-performing machine learning teams, translating complex research into production systems that deliver measurable business impact. Built Viator's AI function from zero to a team of 12+ Principal and Senior Engineers and Scientists; delivered GBP 50M+ across Search, Personalisation, and GenAI. Equally strong on people leadership (career development, 1:1 coaching, OKR facilitation, bottleneck removal) and on hands-on delivery: shipped production LLM systems, agentic AI, real-time ML infrastructure, and shared platform tooling that accelerated the whole org. Published at ACM KDD 2025. Remote-first by default; collaborative in practice.

CORE COMPETENCIES

► **ML Team Leadership:** 0 → 12 team build; hiring; 1:1s; career development; OKR facilitation; bottleneck removal

► **Evaluation & Rigour:** Ablation-first; NLI entailment; BERTScore; G-Eval; trajectory-level agent evaluation

► **Applied AI/ML Delivery:** End-to-end ML lifecycle; GenAI & Agentic AI; LLM orchestration; RAG/GraphRAG; LTR; NLP/ABSA

► **Distributed Systems:** PySpark batch at scale; real-time <50ms P95; multi-region remote team leadership

► **MLOps & LLMOps:** CI/CD eval stacks; model monitoring; cost-aware inference; Arize, Valohai, Seldon, Kubernetes

► **ML Stack:** Python, PyTorch, HuggingFace, LangChain, LangGraph, PySpark, AWS (SageMaker, EMR, Kinesis), Qdrant

► **Strategic Alignment:** CTO/C-suite partnership; product-ML OKR alignment; cross-functional dependency coordination

► **Startup/Scale-up:** High-growth product environment; rapid iteration; quality-velocity balance; iterative delivery

EXPERIENCE

Machine Learning Manager (Head of AI) | Viator – Tripadvisor Group

Mar 2023 – Dec 2025

Remote-first | UK, Europe, US

- Built Viator's ML function from inception: recruited, structured, and led a team of 12+ Principal and Senior ML Engineers and Scientists across NLP, GenAI, Recommender Systems, and Applied Research; designed org structure, team rituals, and career frameworks enabling consistent delivery through a rapidly evolving AI landscape.
- Led team OKR planning, project coordination, and resource allocation across 3+ concurrent workstreams, maintaining strategic alignment with Product and Engineering leadership while keeping engineers focused and unblocked.
- Coached and developed engineers at all levels: ran structured 1:1s, mentored senior contributors on technical influence and stakeholder communication, and redirected work gracefully when priorities or dynamics required it, holding a high bar without eroding trust.
- Delivered ML products with measurable business impact:
 - Personalised Search Ranker V2: two-stage LightGBM + DCN-v2 LTR; +9.5% conversion lift; 47ms P95 latency at global PySpark scale.
 - Agentic Customer Service: GraphRAG knowledge engine, multi-tier memory, HITL escalation, sub-1.5s P95; trajectory-level evaluation; significant Tier-1 deflection.
 - GenAI Content Pipeline: LLM-driven product description system; NLI/BERTScore/G-Eval eval in CI/CD; Responsible AI safety gates; measurable conversion impact.
 - Multilingual ABSA at Scale: opinion clustering across 20+ languages; 80%+ LLM token reduction at marketplace scale.
- Built shared ML platform infrastructure enabling product teams to ship faster and more safely: evaluation framework in CI/CD at PySpark batch scale; Responsible AI safety gate (<10ms latency, 90%+ high-risk incident reduction); MLOps tooling (Arize, Valohai, Seldon); reusable prompt optimisation library published at ACM KDD 2025.
- Reduced cost-per-inference by 27% via LLM orchestration optimisation; published hybrid APE-OPRO at ACM KDD 2025: 18% API cost reduction, matched SOTA on 66-class classification.
- Partnered with CTO and C-suite to align ML investment with company strategy; co-designed GenAI research programme with University of Edinburgh. Total measurable business impact: GBP 50M+.

Lead Data Scientist | ScottishPower
Glasgow, UK

Oct 2020 – Mar 2023

- Led ML delivery for energy forecasting, customer churn propensity, and NILM-based energy disaggregation; built MLOps infrastructure for model reliability under distribution shift in a regulated environment.
- Founded the internal ML Academy: capability-building programme accelerating AI literacy and data-driven decision-making across a large non-technical organisation; demonstrated ability to create ML culture from scratch.

Senior Data Scientist | QueryClick
Edinburgh, UK (remote-first agency)

Sep 2019 – Sep 2020

- Built deep learning multi-touch attribution models (CNN-RNN, LSTM) processing 4M+ daily events; shipped iteratively in a fast-moving environment with hard latency and interpretability constraints; designed evaluation pipeline measuring model stability through a major distribution shift.

Principal Data Science Consultant | Co-libry
Ghent, Belgium | Contract

Jun 2017 – Jul 2018

- Built a real-time personalisation and recommender system in Python for an AI-personalisation startup. Implemented similarity-based and content-based techniques (Jaccard similarity, content embeddings) to improve user-level relevance and conversion.

Senior Data Science Manager, Middle East & North Africa | Nielsen
Dubai, UAE

Jul 2011 – Sep 2016

- Led a 15-member regional ML team across MENA delivering solutions in telecom, retail, and media; introduced structured delivery and QA processes cutting project timelines from 90 to 5 days. Excellence in Leadership Award.

SELECTED DELIVERY

Agentic Customer Service (Viator): End-to-end LLM agent with GraphRAG, multi-tier memory, HITL escalation; trajectory-level + adversarial evaluation; sub-1.5s P95; shipped research-to-production in under 6 months.

Real-Time Personalised Search – Ranker V2 (Viator): LightGBM + DCN-v2 two-stage LTR; +9.5% conversion; 47ms P95 at PySpark global scale.

Cost-Aware APO – ACM KDD 2025: Hybrid APE-OPRO; 18% API cost reduction; 33% faster optimisation; matched SOTA on 66-class classification.

Responsible AI Safety Platform (Viator): Critic-at-the-Edge multi-layer safety gate; <10ms latency; 90%+ high-risk incident reduction; PII policy and source authority hierarchy.

Multilingual ABSA at Scale (Viator): Opinion clustering and sentiment pipeline across 20+ languages; 80%+ LLM token reduction at marketplace scale.

PUBLICATIONS & RESEARCH

- Lead author, ACM KDD 2025: Cost-Aware Automatic Prompt Optimisation, economic optimisation of LLM inference at enterprise scale.
- Co-designed GenAI safety and evaluation research programme with University of Edinburgh.
- Technical writing at snehainair.github.io: ML engineering, LLM evaluation, agent safety, recommender systems.

EDUCATION

M.Sc. Data Science — Lancaster University, UK

MBA, Marketing & Analytics — Lucknow University, India

B.Sc. Chemistry — Mumbai University, India